卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CA-0220 30-B003701S-T0220

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detailed Design DESIGN STAGE

建筑 专业 第 02 卷 第 20 册
Architecture SPECIALITY VOLUME BOOK

卷册名称 检修备品库建筑图（第一次修改）
Volume name Maintenance workshop & spare parts storage(First revision)

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	
图 纸 4 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0220-001	检修备品库建筑图 设计说明 Maintenance workshop & spare parts storage Description of design	B	1		
2	30-B003701S-T0220-002	检修备品库建筑图 平面图 Maintenance workshop & spare parts storage plan	B	1		
3	30-B003701S-T0220-003	检修备品库建筑图 立面图 剖面图 Maintenance workshop & spare parts storage elevation & section	B	1		
4	30-B003701S-T0220-004	检修备品库建筑图 楼梯详图 轻钢屋面详图 Maintenance workshop & spare parts storage detail of the stairde tail of light steel roof	B	1		

注： 本次升版图纸为001、002、003、004图。
The revised drawings are 001、002、003、004.

建筑设计说明

一、项目概况：

项目名称：巴基斯坦默蒂亚里—拉合尔 巴基斯坦默蒂亚里—拉合尔±660kV 直流输电工程/拉合尔±660kV 换流站

建设单位：国家电网/中国电力技术装备有限公司

单体名称	层数	火灾危险性分类	建筑高度	建筑面积	室内外高差
检修备品库	1	丁	16.30	1432.70	0.45

结构形式：钢结构 结构使用年限：50 年，屋面防水等级：1 级耐火等级： 二级

二、建筑施工注意事项：

1. 本工程建筑和装饰用料的选用，规格及色彩应满足设计要求。建筑饰面及装修材料规格和色彩应满足设计要求。
2. 土建施工队、安装施工队装饰等各专项施工队伍间、各专业间均应密切配合，施工或安装前先应全面清楚了解有关工种设计图纸内容要求（包括基础结构部份）。发现设计问题或现场有变化情况时，应及时与设计人员联系，协商确定解决方法，进而确保工程进度与施工质量。
3. 本项目工程建筑关键部分用料，如油漆和涂料等的规格和质量等要求，须经监理确认无误，以确保工程质量。
4. 建筑平面所注尺寸均为结构尺寸。

三、建筑消防设计：

1. 单体间间距均大于 9m，满足消防间距要求。
2. 钢结构防火要求如下：

序号	构件名称	耐火极限（h）	防火涂料类型	备注
1	柱、柱间支撑	2.00	厚涂型防火涂料	
2	梁、桁架、设备吊梁	1.50	薄涂型防火涂料	
3	屋盖承重构件、屋面支撑、系杆	1.00	薄涂型防火涂料	

四、建筑施工说明

1.墙体

1.1 概述

1.1.1 本项目砌块采用当地标准机制粘土烧结砖，砖长度为 230，砌块和砂浆强度详结施。

1.1.2 从基础向上砌筑的墙体在标高-0.060 处设置 1 层 20 厚 1:3 水泥砂浆(掺 5%防水剂)的防潮层。(该部位有结构圈梁则可以取消墙体防潮层)

1.1.3 外粉刷必须设置分格缝。

1.1.4 建筑物顶层和底层应设置通长现浇钢筋混凝土窗台梁，高度不宜小于 120mm，配筋详结施；其他层在窗台标高处应设置通长现浇钢筋混凝土板带。窗口底部混凝土板带应做成里高外低；房屋两端顶层砌体沿高度方向应设置间隔不大于 1.3m 的现浇钢筋混凝土板带。板带的配筋及混凝土强度详结施。

1.1.5 在两种不同基体交接处，应采用钢丝网抹灰或耐碱玻纤网布聚合物砂浆加强带进行处理，加强带与各基体的搭接宽度不应小于 150mm。顶层粉刷砂浆中宜掺入抗裂纤维。

1.1.6 室内墙、柱面、门窗的阳角，在 2000 高度内均做 1:2 水泥砂浆护角线，每侧宽度不小于 50。

1.1.7 外墙窗台、窗楣、雨篷、阳台压顶和突出腰线等上面应做流水坡度，且不应小于 25，下口应设滴水槽，深度为 10mm，宽度为 10-12mm，距水平板外缘 30mm 处或离墙面 30mm 处设置断水口，滴水槽内涂水柏油。

1.1.8 顶层和底层应设置通长现浇钢筋混凝土窗台梁高 200mm，配筋详结施。其它层在窗台标高处设置钢筋混凝土板带；房屋两端顶层应设置通长现浇混凝土板带，间隔 1.3m。板带 C20，高 80mm，配筋详结施。

1.1.9 内墙墙面粉刷层满铺抗裂钢丝网。外墙与成品电缆柜接触部分为混凝土实心砖砌筑。如遇混凝土墙面，先刷素水泥浆（加 3%建筑胶）一道，满批腻子打光，再做外墙饰面。勒脚墙砖吸水率 E≤3%，破坏强度≥1300N. 可根据实际砖铺贴尺寸，做调勒脚高度，尽量采用整砖。

1.1.10 内墙乳胶漆耐洗刷性（次）≥1000，VOC 含量≤100g/L；瓷砖吸水率不大于 6%，破坏强度不小于 600N；

1.1.11 室内窗台贴黑色人造石，人造石突出墙面约 20mm，沿口倒圆角。窗台板

1.2 墙体构造做法

1.2.1 外墙 1 做法(用于勒脚处)：

1:1 水泥砂浆掺粘结剂(或专用粘结剂)粘贴深灰色面砖(配套填缝剂勾缝)

20 厚 1:3 水泥砂浆找平

砌块墙体作基层处理

1.2.2 外墙 2 做法(用于勒脚以上)：

丙烯酸外墙涂料面漆两遍

满涂底漆一道

外墙腻子批平

20 厚 1:3 水泥砂浆找平

不同材料间用钢丝网片铺贴

砌块墙体基层处理

1.2.3 内墙做法：

A 级白色无机涂料一底二面

内墙腻子三遍

6 厚 1:0.5:2.5 混合砂浆抹平

9 厚 1:0.5:3 混合砂浆扫毛

墙体非同一材料时接缝处需增加钢丝网片

素水泥浆 1 道

基层处理

2．地面

（环氧砂浆耐磨地坪）地面构造做法

环氧耐磨地坪（按厂家工艺流程）；

50 厚 fc’=28MPa 细石混泥土内配 Φ10@200 双向钢筋，每 6×6m 设分仓缝，内填胶泥，遂捣遂磨平；

20 厚 1:3 水泥砂浆；

1.5 厚聚氨酯防水层两道；

150 厚 fc’=21MPa 混凝土垫层，含水率≤6%；

素土夯实压实系数不小于 0.94。

3．屋面工程：

3.1 概述

选用的是优质压型钢板屋面系统，主要包括：防水、保温、抗风等方面。

3.2 构造

1、压型钢板屋面构造层（从上向下）

a. 屋面板：0.8mm 厚 65/400 型镀铝锌压型钢板屋面板，颜色绿色

b. 防水层:0.49mm 厚防水透气膜

c. 保温层：2X50mm 厚离心玻璃棉

d. 底板层：0.6mm 厚镀铝锌压型钢板

e. 檩条：详见结构图纸

f. 原钢主体结构：详见结构图纸

3.3 节点设计

在屋脊的处理上，选用与板型相吻合的铝合金配套密封件与屋面板板肋用防水铆钉连接固定，并在密封件后塞入与板型相一致的屋脊泡沫密封条，然后将屋脊盖板与密封件用铆钉在中间固定。

3.4 抗风设计

3.4.1. 铺板方向

在考虑抗风设计的时候，首先应考虑铺板的方向，在工程所在地施工时，该地区的主导风向在哪个方向，铺板时应让大边部分背对风向。

3.4.2. 固定方式

本方案首先将固定支座与檩条固定，然后将屋面板扣在固定支座的梅花头上，再用电动锁边机将板的大小肋锁在固定支座上。采用 360 直立锁边。

4.油漆涂料

4.1. 防腐涂料的底漆和面漆应相互配套。

4.2. 涂装时环境温度应在 5~38° C 间，相对湿度不应大于 85%。

4.3. 构件表面有结露时不得涂装，涂装后 4h 内不得淋雨。

4.4. 钢结构构件在涂刷防锈蚀涂料前必须将构件表面的毛刺、铁锈、油污及附着物清除干净,除锈方法采用喷砂抛丸除锈，等级应达到 Sa2.5。

4.5. 底漆及面漆材料应配合建筑防火规定：一般应涂装 4~5 遍

室外构件涂层配套漆膜干膜总厚度不低于 150um

室内构件涂层配套漆膜干膜总厚度不低于 120um

5.其它

5.1 防雷设施:详电施图纸

5.2 建筑图纸中各专业综合留洞示意图后出，电气、水工、暖通等专业留孔大小及位置详见各专业图纸（以各专业图纸为准）。

5.3 任何图纸上有矛盾与疑问之处，要在着手工作之前报请建筑师和工程师澄清如出现中英文版本不一致情况，以中文版为准。

Project Narrative

1.General

Project name：±660kV HVDC Project from Matiari to Lahore in Pakistan

Development organization：State Grid / China Electric Power Equipment And Technology CO .TD

Name	Story	classification of fire hazards	Height	Area	Altitude intercept
Maintenance workshop & spare parts storage	1	the fouth of the ten Heavenly Stems	106.30	1432.7	0.45

Structural style: steel structure　　Service life of structure: 50 years

Roof waterproofing grade: I　　Fire resistance class 2

2. Precautions of construction

1) The selection, specifications and color of the construction and decoration materials should meet the design requirements.

2) Different majors of construction teams such as construction team and installation team should cooperate with each other closely. Construction teams should have a comprehensive understanding of the relevant blueprints before construction or installation. When discovering the design mistake or changes during the construction, they should contact the design unit in time to discuss the solution in order to ensure the progress and quality of the project.

3) The specification and quality of material requirements of the key parts such as paintings should be confirmed by supervisor in order to ensure the quality of the project;

4) The dimensions of the building plan are structural size.

3: Architectural fireproof design

1) In order to satisfy the requirement of separation distance for fireproof, the distance between two single buildings should be larger than 9m;

2) Requirements for fire protection of steel structure are as follows

NO.	Component name	Fireresistance limit (H)	Types of fire retardant coatings	Remark
1	Column and column support	2.00	Heavily coated fire retardant paint	
2	Beam, truss, equipment hanging beam	1.50	Thin coating type fire retardant coating	
3	Roofbearing member, roof support and tie rod	1.00	Thin coating type fire retardant coating	

4: Construction instructions

1. Wall

1.1 General

1.1.1 The masonry structure of this project uses local clay brick, The brick length is 230 and the standard strength consult structure design drawings.

1.1.2 The walls built from the foundation should set one layer of moisture proof layer at 0.060m level high which is 20mm width and using 1:3 cement mortar (with 5% water-proof agent). (Moisture proof layer can be canceled if beam wall)

1.1.3 Exterior paint must be provided with slit.

1.1.4 Top and bottom of building should be set long cast-in-place concrete sill beam, the height of beam is less than 120mm,the set of steel bar refer to the detail of structure; other layers of floor should be set reinforced concrete slab continuous cast-in-situ belt at the window. The outside, the top masonry at both ends of the house shall be provided with a cast-in-place reinforced concrete slab with an interval of not more than 1.3m along the height. Structural slab reinforcement and concrete strength consult structure design drawings.

1.1.5 the steel bar must be embedded between the balcony and the main part of the building.

1.1.6 Indoor wall, doors and windows, cylindrical corner under 2000mm height are 1:2 cement mortar protection line, the width of each side no less than 50mm.

1.1.7 At the part of windowsill, window lintel, awning, balcony top and belt line should set water slope, and should not be less than 25mm, at the bottom there should set drip line slot. The depth is 10mm, the width is 10-12mm, and from the wall 30mm set a water outlet, brushing bit mastic solution into the drip line slot.

1.1.8 On the top and bottom should be set long cast-in-place reinforced concrete beams in the window, the set of steel bar refer to the detail of structure, totally beam is 120mm high. The other layers in the window at the elevation reinforced concrete strip; both ends of the top housing should be set through the concrete slab with interval 1.3m. Structural slab reinforcement and concrete strength consult structure design drawings.

1.1.9 Iinterior wall's painted layer covered with anti-crack steel wire mesh. The contact part of the external wall and the cable cabinet is made of concrete solid bricks. At the part of concrete wall, there should brush cement paste (3% building glue) at first, then putty, then do exterior finishes. Water absorption rate of plinth's material should be less than 3%(E < 3%), damage strength > 1300N. According to the actual size of tiles, fine-tuning the plinth height, as a whole brick.

1.1.10 Interior wall latex paints scrub resistance (time) should be more than 1000, VOC content is less than 100g/L; tile's water absorption is not more than 6%, the failure strength is not less than 600N.

1.1.11 Indoor windowsill use black artificial stone, artificial stone prominent wall about 20mm, fillet along the corner. The combination of windowsill and wall is sealed with white silica gel.

1.2 Walls

1.2.1 Exterior wall 1 (For the plinth):

Dark gray tiles (using relevant joint mixture) are pasted by 1: 1 cement mortar mixed with binder (or special binder);

Smoothed by 20 mm and 1: 3 cement mortars;

Add steel wire mesh to the joints of non-identical materials of the wall;

Treat the masonry walls at the base level;

The decoration layer for surface on interior wall;

1.2.2 Exterior wall 2 (Above the plinth):

Acrylic exterior paint finish twice

Apply a full primer

Smooth the putty on exterior wall;

20 mm and 1: 3 cement;

Wire mesh is used to cover different wall materials

Treat the base level for masonry walls;

1.2.3interior wall :

Synthesized resin latex coatings：Once primer, twice pastry;

Paint putty on interior wall for three times;

Smoothed by 6 mm 1:0.5:2.5 mixed mortar;

Flatted by 9 mm 1:0.5:3 mixed mortar;

Add steel wire mesh to the joints of non-identical materials of the wall;

Paint one layer of simple grout;

Treat the base level;

2. Flooring

Floor construction practices

Rpoxy wear-resisitant floor(According to the manufacturer's process)

50mm fc'=28MPa fine stone concrete with φ10@200 bidirectional steel bar, each 6X6m is divided into seams , filled with clay, stir it while stirring

1:3 cement mortar with thickness of 20mm

Two layers of 1.5mm polyurethane waterproof

150mm fc' = 21MPa concrete cushion, moisture content ≤ 6%

The soil compaction coefficient of plain soil is not less than 0.94

3. Roof

3.1 General

The high-quality profiled steel roofing system is selected, including: waterproof, thermal insulation, wind resistance and so on

3.2 Roof construction practices

Profiled steel sheet roof structural layer (from top to bottom):

a. Roof panel: 0.8mm thick aluminium zinc profiled steel sheet roof panel, green color
b. Waterproof layer: 0.49mm thick waterproof breathable film
c. Insulation layer:2X50mm thick Centrifugal insulation cotton
d. Bottom Roof layer: 0.6mm thick aluminum zinc profiled steel sheet
e. Purlin: refer to structural design drawings
f. Main steel structure: refer to structural design drawings

3.3 detail node design

In the ridge of roof processing, aluminum alloy selection and type waterproof rivets, foam and seal in squeezed and profile consistent with the seal, and then the roof plate with a rivet in the middle of the fixed and seal.

3.4 Wind resistant design

3.4.1 Laying plate direction

When considering the wind resistant design, should first consider the laying plate direction, the location of the project construction, the area of the dominant wind direction in which direction should make planking part of the back side of the wind

3.4.2 Fixed mode

First fixed clamped column and purlin, then fixed the roof panel on the clamped column, at last use the electric sewing machine to put the rib locks clamped column together

4. Paint

4.1 Primer and finish painting of anticorrosive paint should be matched with each other

4.2 The ambient temperature should be between 5~38C, and the relative humidity should not be greater than 85% during painting

4.3 When the surface of the component has dew, it shall not be painted, and no rain in the 4 hours after painting

4.4 The steel structure component must clean the burr, rust, oil stain and adhesive on the surface of the component before brushing the antirust coating. The rust removing method adopts sand blasting and rust removing, and the grade should be Sa2.5.

4.5 Primer and topcoat materials should be in accordance with the building fire protection regulations: generally should be painted 4~5 times

The total dry film thickness of the exterior component painting is not less than 150um

The total thickness of dry film for interior component painting is not less than 120um

5. Else:

5.1 Lightning protection facilities: refer to detail of electrical drawings

5.2 Electrical, hydraulic, HVAC and other specialty's reserve holes, refer to relevant drawings.

5.3 If there are any contradictions on any drawings, contact the designer before construction. In the event of a difference between the Chinese and English versions, the Chinese version shall prevail.

Reference Drawings

Drwg. No.	Title
30-B003701S-T0248	建筑通用详图 General details of the building

Revisions				
	A	刘鹏	2017.11.30	第一版 First Edition
		Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - E C I - D D - C A - 0 2 2 0 - 0 0 1 B

	姓名 Name	日期 Date
设计 Designed	刘鹏	2019.09.06
校核 Checked	郭春	2019.09.06
审核 Reviewed	刘鹏	2019.09.06
批准 Approved	黄奇	2019.09.06

检修备品库建筑图

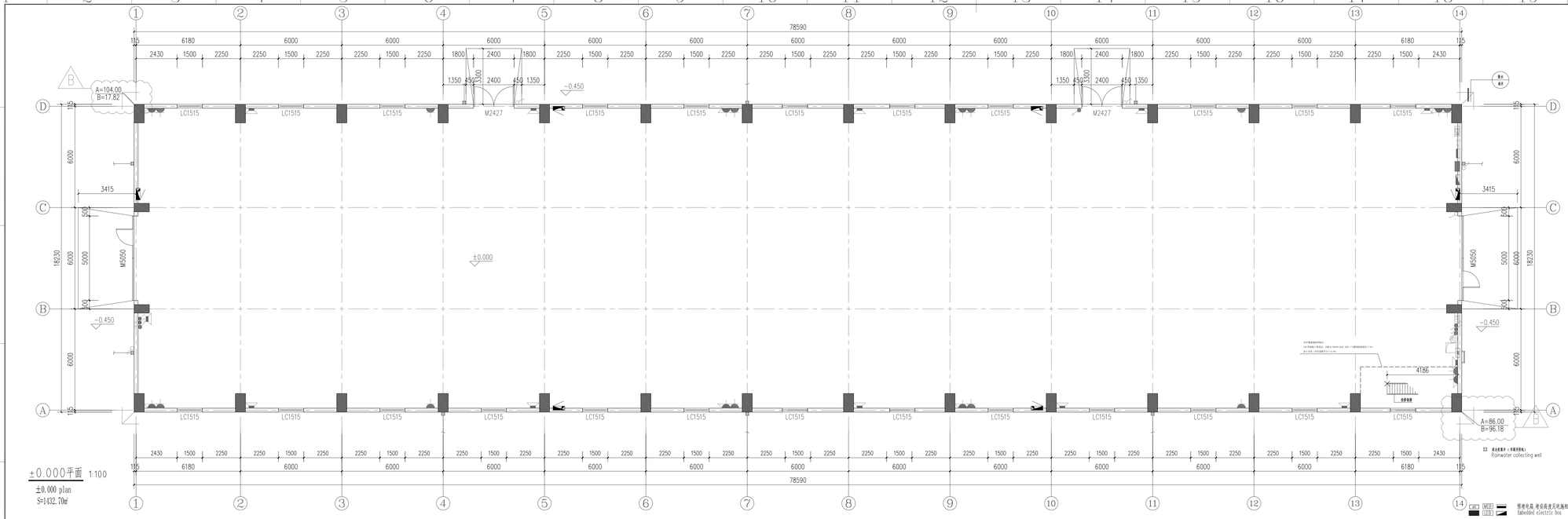
设计说明

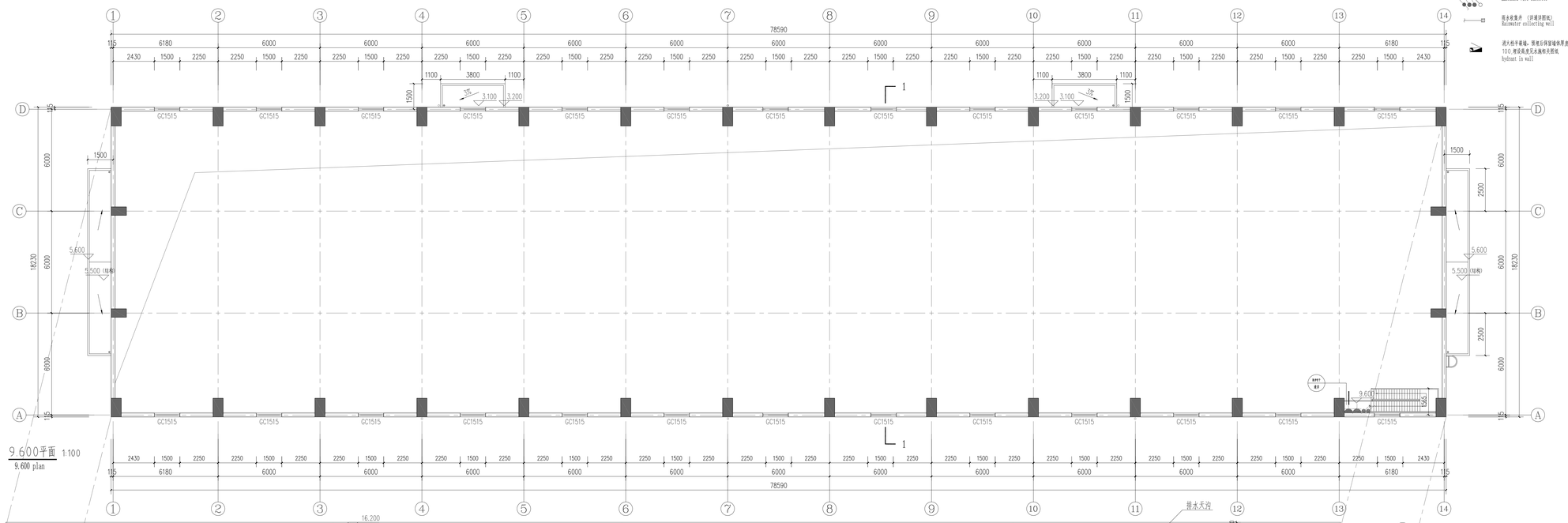
Architectrual drawing of Maintenance workshop & spare parts storage

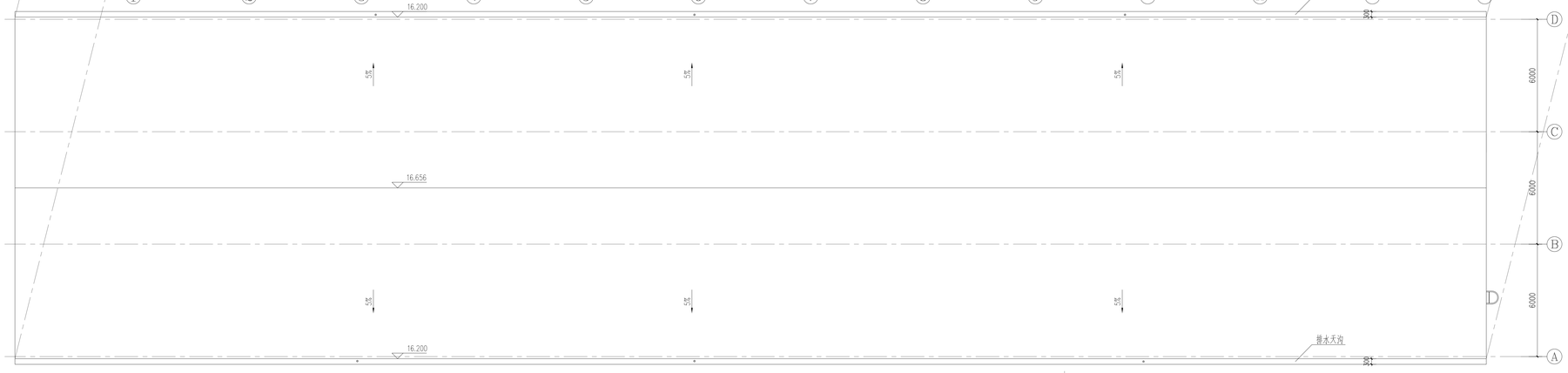
Description of design

比例 Scale: - -　图号 DWG NO.: 30-B003701S-T0220-001　版本 REV.: B

594X841





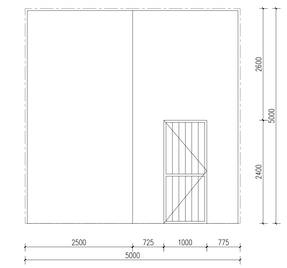


预埋电箱,埋设高度见电施相关图纸
Embedded electric box

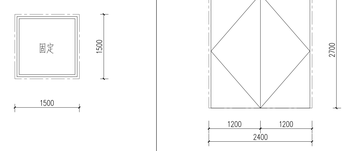
预埋线槽盒,埋设高度见电施相关图纸
Embedded wire cassette

雨水收集井（详通详图纸）
Rainwater collecting well

消火栓平面墙，预埋后保留墙体厚度100,埋设高度见水施相关图纸
Hydrant in wall

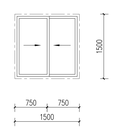


编号 number	M5050	洞口尺寸 door hole size	5000X5000
备注 remarks	钢板电动移门 steel plate moves the door	数量 number	2



编号 number	GC1515	洞口尺寸 door hole size	1500X1500
备注 remarks	铝合金窗 aluminum alloy window	数量 number	26

编号 number	M2427	洞口尺寸 door hole size	2400X2700
备注 remarks	钢板门 steel plate door	数量 number	2



外窗断桥隔热铝合金窗框，窗玻璃采用普通钢化玻璃6+12A+6，所有可开启窗附纱窗。

Outside the window of aluminum alloy of the heat insolation that break the window frame, window using ordinary tempered glass 6+12+6, all of the windows which can open should install the screen window .

编号 number	LC1515	洞口尺寸 door hole size	1500X1500
备注 remarks	铝合金窗 aluminum alloy window	数量 number	24

门窗详图 1:50
Windows and Doors detail

Reference Drawings

Drwg. No.	Title
30-B003701S-T0268	建筑通用详图 General details of the building

Revisions				
A	Name	2017.11.30 Date	第一版 First Edition Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV UHDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE 施工图 Construction Drawing

M L H V D C - L C S - E C I - D D - C A - 0 2 2 0 - 0 0 2 B

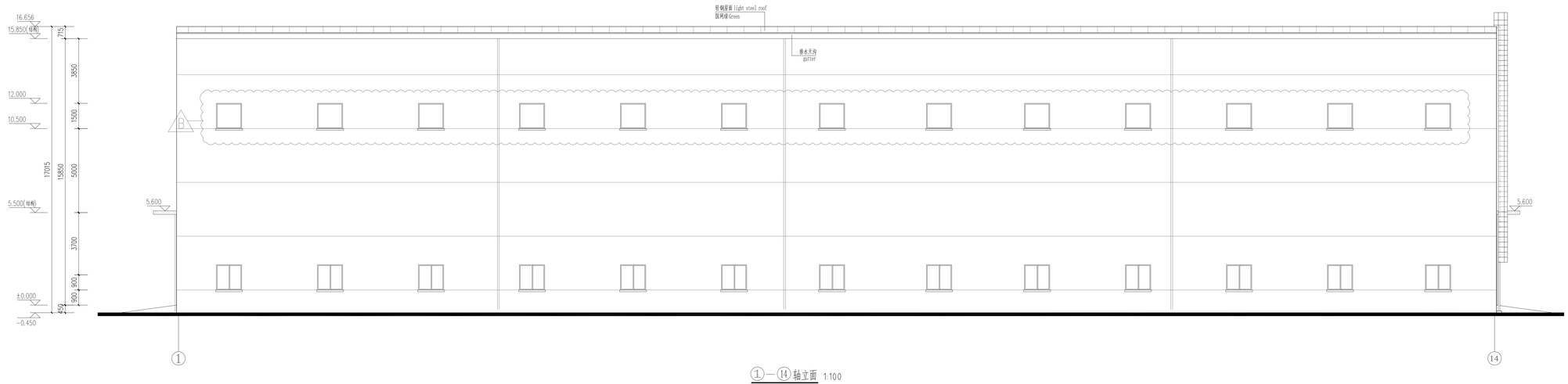
	姓名 Name	日期 Date
设计 Designed		2019.09.06
校核 Checked		2019.09.06
审核 Reviewed		2019.09.06
批准 Approved		2019.09.06
比例 Scale	1:100	图号 DWG NO.

检修备品库建筑图
平面图
Maintenance workshop & spare parts storage plan

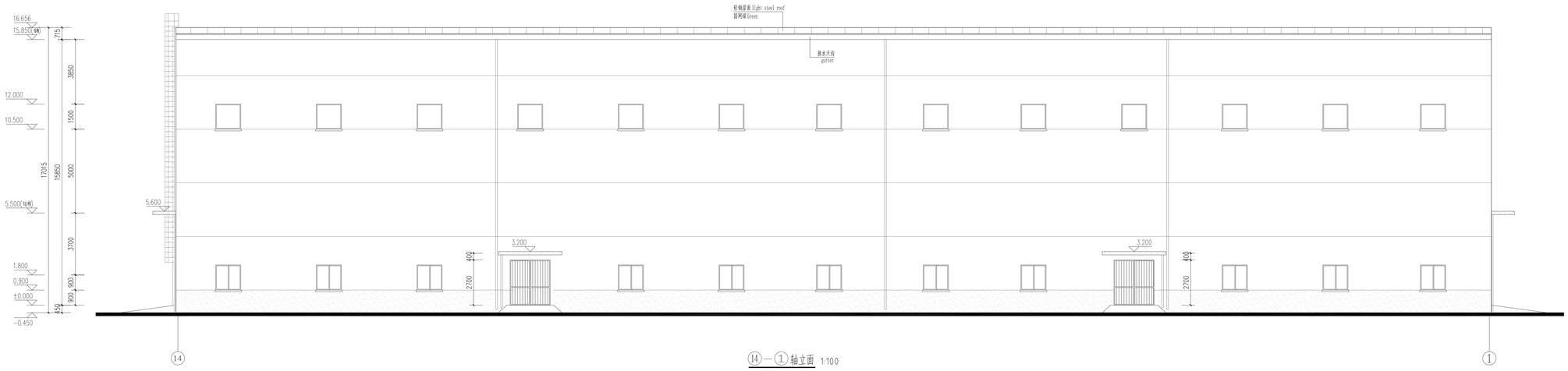
30-B003701S-T0220-002

版本 REV. B

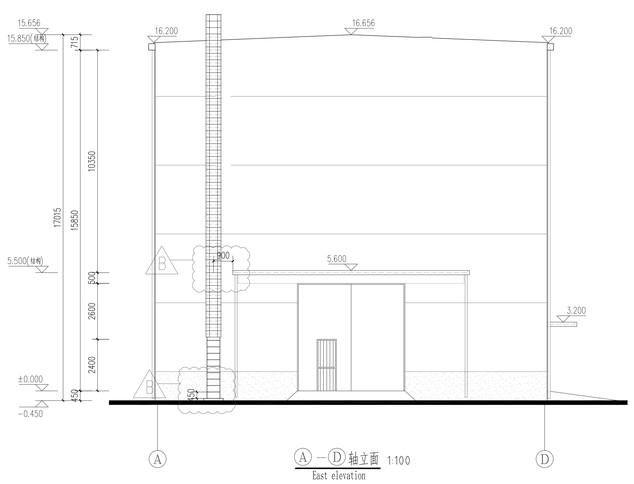
专业 Speciality	会签 Sign	日期 Date
结构 Structure		2019.09.06
电气 Electrical		2019.09.06
暖通 HVAC		2019.09.06
给排水 Water supply & drainage		2019.09.06



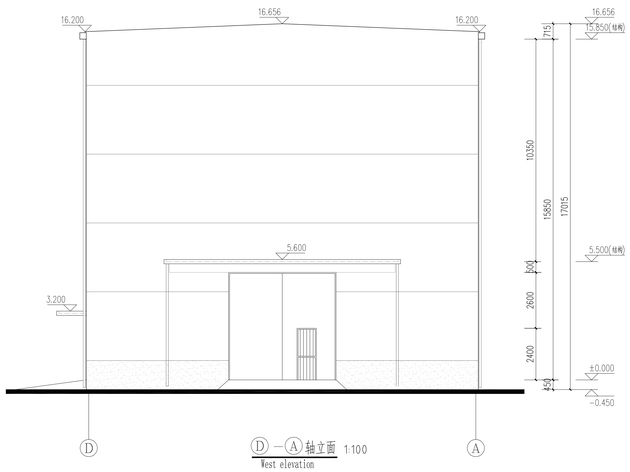
①—⑭轴立面 1:100
South elevation



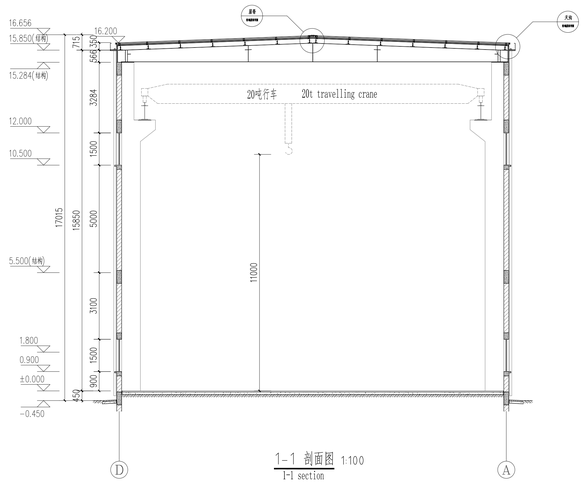
⑭—①轴立面 1:100
North elevation



Ⓐ—Ⓓ轴立面 1:100
East elevation



Ⓓ—Ⓐ轴立面 1:100
West elevation



1-1 剖面图 1:100
1-1 section

Reference Drawings

Drwg. No.	Title
30-B003701S-T0288	建筑通用详图 General details of the building

Revisions

	Name	Date	Notes
A		2017.11.30	第一版 First Edition

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
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CONTRACTOR: 国家电网 STATE GRID 中国电力技术装备有限公司

SUBCONTRACTOR: East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

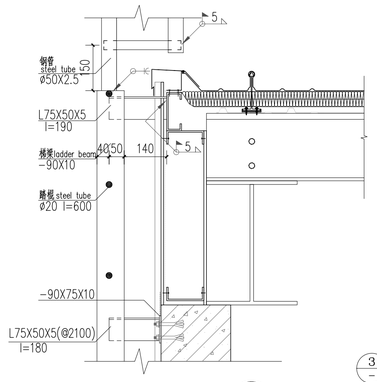
设计阶段 PHASE: 施工图 Construction Drawing

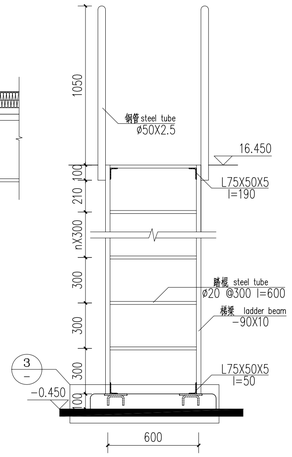
M L H V D C - L C S - E C I - D D - C A - 0 2 2 0 - 0 0 3 B

	姓名 Name	日期 Date
设计 Designed		2019.09.06
校核 Checked		2019.09.06
审核 Reviewed		2019.09.06
批准 Approved		2019.09.06

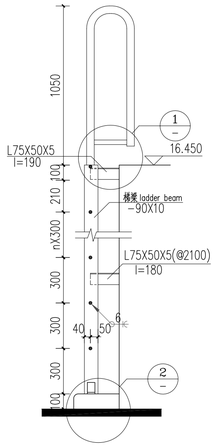
检修备品库建筑图
立面 剖面
Maintenance workshop & spare parts storage
elevation & section

比例 Scale: 1:100
图号 DWG NO.: 30-B003701S-T0220-003
版本 REV.: B

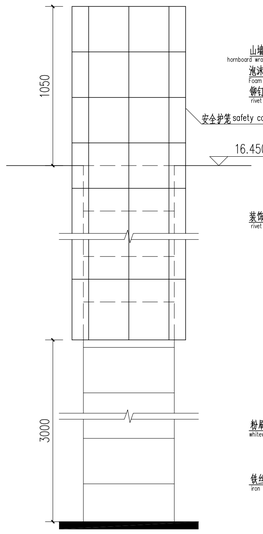




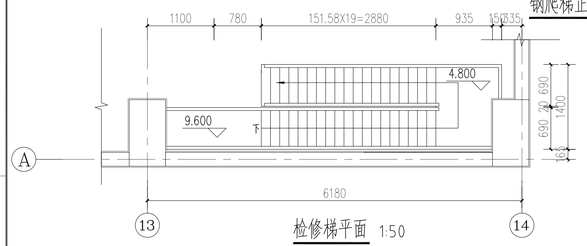
钢爬梯正立面 1:20



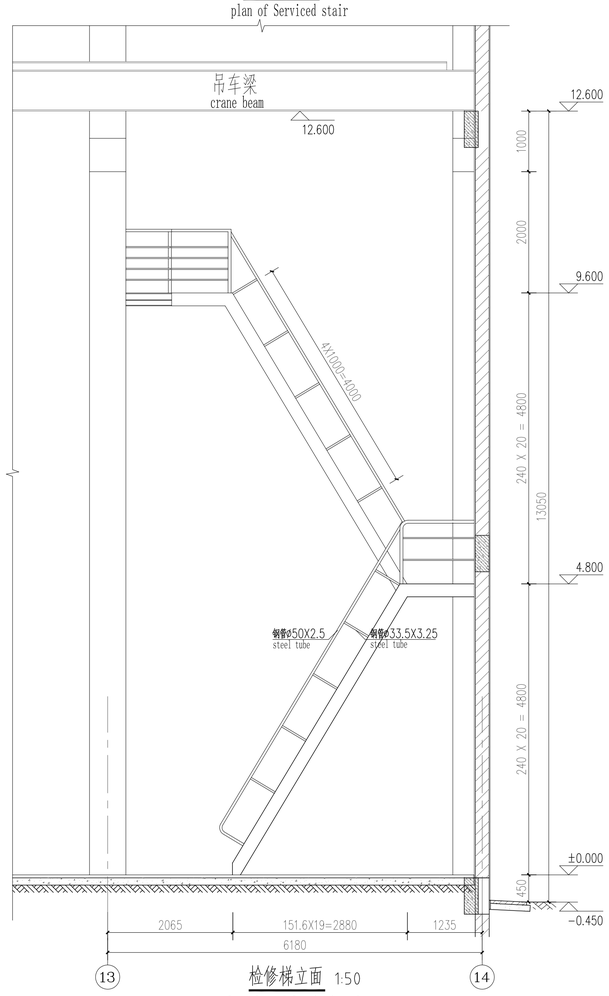
钢爬梯侧立面 1:20



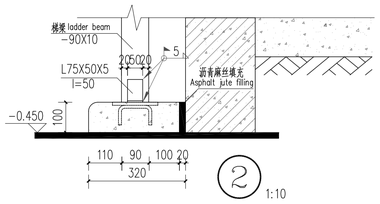
钢爬梯示意 1:20

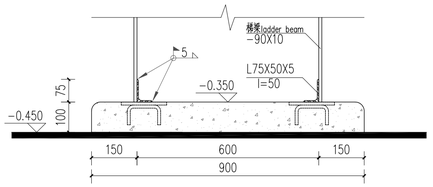


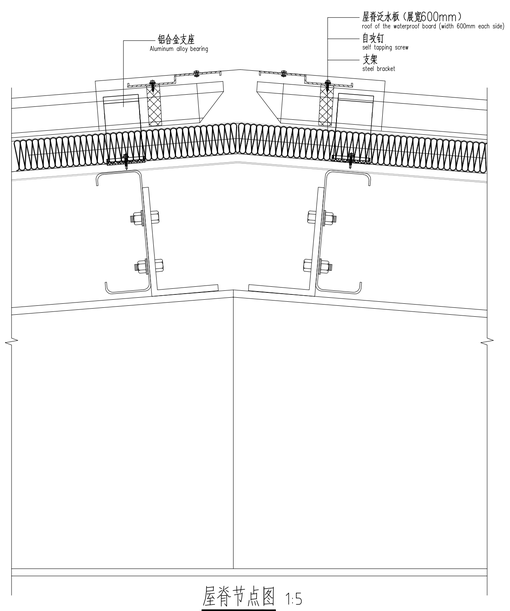
检修梯平面 1:50
plan of Serviced stair



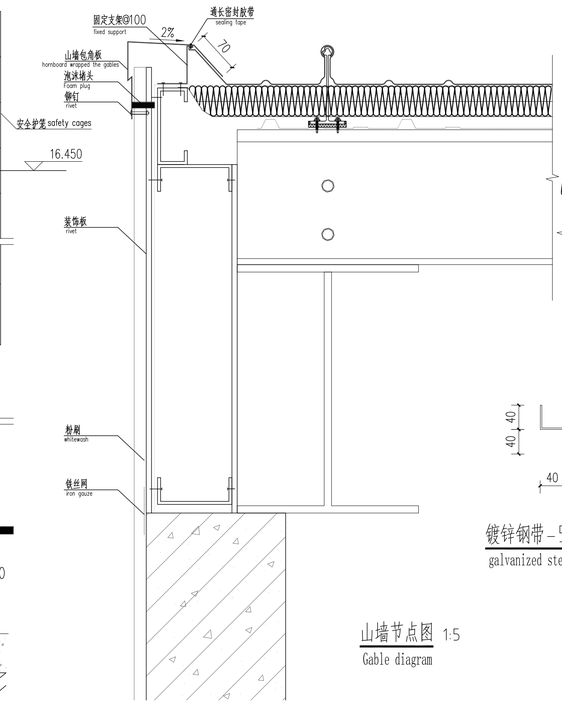
检修梯立面 1:50
elevation of Serviced stair



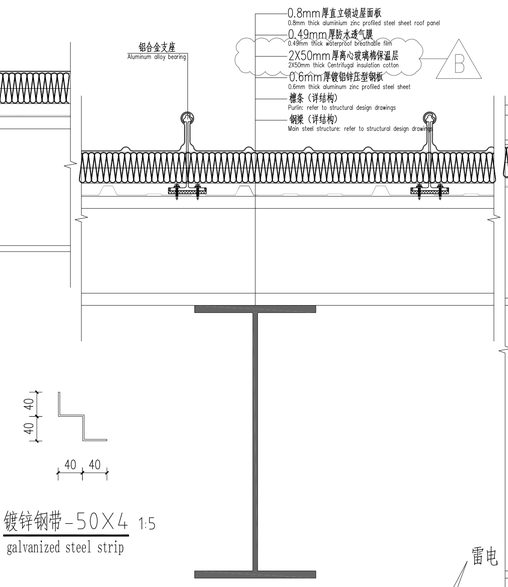




屋脊节点图 1:5
crest node diagram

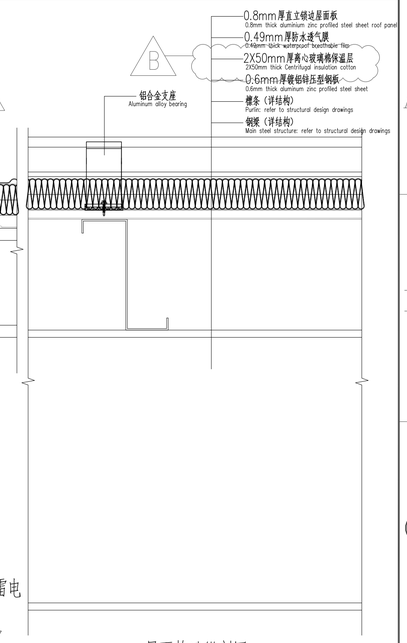


山墙节点图 1:5
Gable diagram

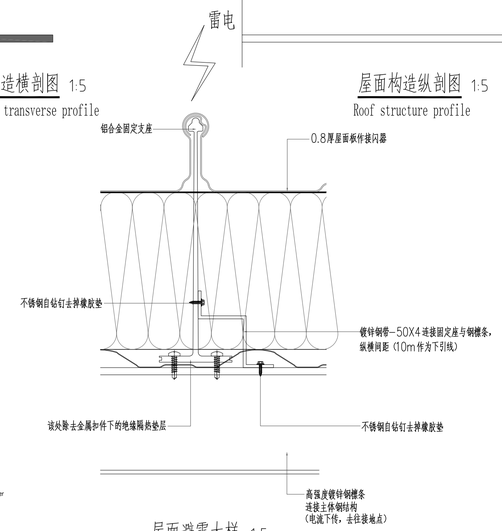


镀锌钢带-50X4 1:5
galvanized steel strip

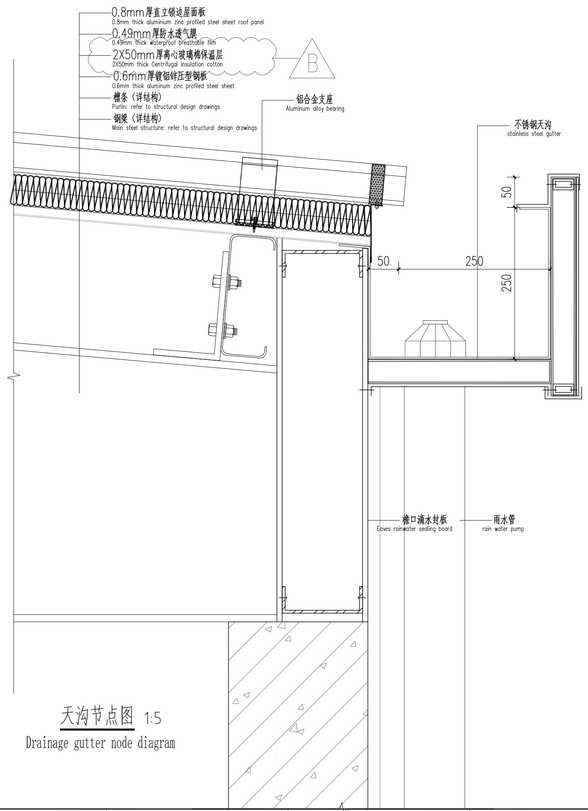
屋面构造横剖图 1:5
Roof structure transverse profile



屋面构造纵剖图 1:5
Roof structure profile



屋面避雷大样 1:5
Lightning protection facilities detail



天沟节点图 1:5
Drainage gutter node diagram

Reference Drawings	
Drwg. No.	Title
30-B003701S-T0248	建筑通用详图 General details of the building

Revisions

	Name	Date	Notes
A		2017.11.30	第一版 First Edition

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

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±660kV Lahore Converter Station

设计阶段 PHASE 施工图 Construction Drawing

M L H V D C - L C S - E C I - D D - C A - 0 2 2 0 - 0 0 4 B

	姓名 Name	日期 Date
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校核 Checked		2019.09.06
审核 Reviewed		2019.09.06
批准 Approved		2019.09.06

检修备品库建筑图
楼梯详图 轻钢屋面详图
Maintenance workshop & spare parts storage
detail of the stair detail of light steel roof

比例 Scale	1:50	图号 DWG NO.	30-B003701S-T0220-004	版本 REV.	B

594X841